Kalp Shah

Roll No.:24110152
B.Tech - Electrical Engineering
Minor in Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology, Gandhinagar	8.41/10 (Current)	2024-2028

Experience

- **AIResQ ClimSols** — *May 2026 - Jul. 2026*
  Machine Learning and Computer Vision Research Intern — Gandhinagar, India
  – Engineered a Computer Vision pipeline to **estimate Flood Depth** from CCTV Footage in Agartala in **4 seconds**
  – Trained **YOLO** Models to detect flood level markers painted on light poles with **mAP@50 of 0.78**
  – Extracted Flood waterline from Images using **Semantic Segmentation with SAM2** and **Edge Detection**
- **ABSix Robotics** — *May. 2025 - Jul. 2025*
  SWE Intern — Gandhinagar, India
  – Engineered a **cross-platform desktop application (Python, Qt)** with real-time robot-student communication via the **Dynamixel SDK** deployed across 15+ challenges and actively used by **100+ students**
  – Integrated a **RAG pipeline (LangChain + ChromaDB)** into the platform for students to query robotics concepts

Projects

- **Controllable Font Generation using cGANs**
  Personal Project — Github
  – Engineered a conditional dataset from **281,000+ TMNIST images**, enabling precise multi-label typographic control by parsing implicit font nomenclature into explicit binary style vectors
  – Architected a PyTorch Conditional **Generative Adversarial Network (cGAN)**, modeling a 784-dimensional feature space by concatenating latent noise tensors with one-hot character and style condition vectors during training
  – Synthesized novel, cohesive 94-character font families, ensuring consistent stylistic features across the generated alphabet by locking latent noise vectors during inference achieving an **FID Score of 25.7466** over 1000 generations
- **CPU Ray Tracing Engine in C++**
  Personal Project — Github
  – Implemented a physics-based CPU ray tracer in **C++** from scratch, rendering scenes with hundreds of spheres
  – Modelled diffuse, metallic, and dielectric surfaces with recursive ray bouncing for realistic light simulation
  – Integrated multi-sample anti-aliasing and optimized vector math, achieving a render time of **0.9s**
- **Realtime Carbon Credits Market Analysis Platform** — *Nov. 2025 - Dec. 2025*
  Inter-IIT Tech Meet 14.0 - Pathway, High Preparation Problem Statement — Github
  – Architected a full-stack real-time analytics platform (**Next.js, WebSocket, PostgreSQL**) serving live carbon credit pricing, trend signals, and risk analysis to end users as part of an **8-person team**
  – Built a data streaming pipeline using **Kafka** and **Debezium** (CDC) to capture live **PostgreSQL** changes, cutting data freshness latency from **60 minutes to under 3 seconds**
  – Developed a multi-agent **RAG system (LangChain, DeepSeek)** with role-separated agents for retrieval, reasoning, and synthesis generating context-aware investment recommendations
- **Landing Location Detection Pipeline for Mars Rover using Classical Computer Vision** — *Jan. 2026 - Apr. 2026*
  Prof. Shanmuganathan Raman, IITGn — Github | Research Poster
  – Developed a Computer Vision pipeline to identify **safe landing zones on Mars** with **ISRO** under compute constraints
  – Designed a modular system utilizing Sobel derivatives for **slope detection**, Fast Fourier Transforms for **texture analysis**, and the Maximal Rectangle Algorithm for **zone identification**
  – Achieved a low computational power requirement solution with **1.5 seconds runtime** on consumer-level hardware

Achievements

- **Expert on Codeforces**, Handle: *kalpshah1808*, Max Rating: 1785, **India Rank ∼430** out of 98000 — *2026*
- **Integral Cup Round 1 Rank 47**, among 2500+ participants, sponsored by **Optiver** and **Da Vinci Trading** — *2026*
- **AMS Derive Round 1 Rank 150**, among 2500+ participants, sponsored by **Jane Street** and **QRT** — *2026*
- **JEE Advanced 2024**, Secured **AIR 6637** among 0.18 million candidates appearing for the test — *2024*
- **JEE Mains 2024**, Secured **AIR 3917** among 1.4 million candidates appearing for the test — *2024*

Skills

- **Languages**: Python, C/C++, SQL, HTML/CSS, Javascript, Assembly, Verilog
- **ML/CV**: PyTorch, OpenCV, scikit-learn, Langchain, Pandas, Numpy, Matplotlib
- **Backend/Tools**: Node.js, WebSocket, PostgreSQL, Kafka, Debezium, Docker, REST APIs, Git, CMake
- **Frontend**: React.js, Next.js, Tailwind CSS